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# Distribution, nature and threats to soils of the Australian Alps: A review

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**Abstract** The Australian Alps contain an assemblage of soil types that is unique on the Australian continent. The above-ground ecosystems of the Australian Alps have received considerable scientific attention but research relating to the nature of its soils has been much more limited. A fuller understanding of the role of soils in these ecosystems is required to inform effective management strategies. This review was undertaken to assess existing research on soils in the Australian Alps. We aimed to summarise our current knowledge of their nature, distribution and characteristics, to examine the services they provide and to assess their vulnerability to the range of threats that exist to the soil resource both local and external. Soils of higher elevations, namely Transitional Alpine Humus Soils, Alpine Humus Soils and upland Peat Soils are particularly important to the ecology, hydrology and potential carbon storage of the region, yet our understanding of the nature, formation and functioning of these soil types remains weak. A series of knowledge gaps and research priorities are identified, relating to basic knowledge needs on the formation, distribution and function of these soils, particularly their microbial populations and the impacts of specific threats (i.e. climate change, grazing, fire, visitors, infrastructure, feral animals and pollution).

**Key words:** alpine humus soils, Australian Alps, research priorities, soils, threats.

## INTRODUCTION

The Australian Alps bioregion (OEH 2019; DAWE 2021a, b) is a mountainous area in south-eastern Australia covering an area of 12 330 km$^2$ in New South Wales (NSW), Victoria (Vic), and the Australian Capital Territory (ACT) (Fig. 1). The Australian Alps contain the Australian mainland's only true 'alpine' zones (Körner *et al.* 2011), with 7938 km$^2$ above the climatic tree limit, of which 54% is in NSW, 36% is in Victoria and 10% is in the ACT (DPIE 2020). The Alps contain Australia's only peaks exceeding 2000 m elevation, with the maximum elevation of 2228 m at Mount Kosciuszko. Much of the Australian Alps bioregion is protected within a group of 11 national parks, most notably Kosciuszko National Park (NP) (NSW), Alpine NP and Snowy River NP (Vic) and Namadgi NP (ACT). The Kosciuszko Biosphere Reserve is recognised by the United Nations Educational, Scientific and Cultural Organization (UNESCO), and the alpine humus soils in particular are considered to be of international significance (NPWS 2014).

The Australian Alps contain landscapes and soils that are unique in Australia due to their relatively high elevation, cool wet climate and consequently high rates of organic matter accumulation (Costin *et al.* 1952; Costin 1954). Soils over much of the Australian Alps are deep compared with similar environments internationally, resulting from lengthy weathering and organic matter accumulation on mostly moderately inclined slopes of an uplifted surface (Costin *et al.* 1952; Costin 1954, 1955). Soils in the Australian Alps tend to become more organic with elevation and are therefore more comparable to temperate 'montane' zones globally (Djukic *et al.* 2010; Chen *et al.* 2016; Ma & Chang 2019), rather

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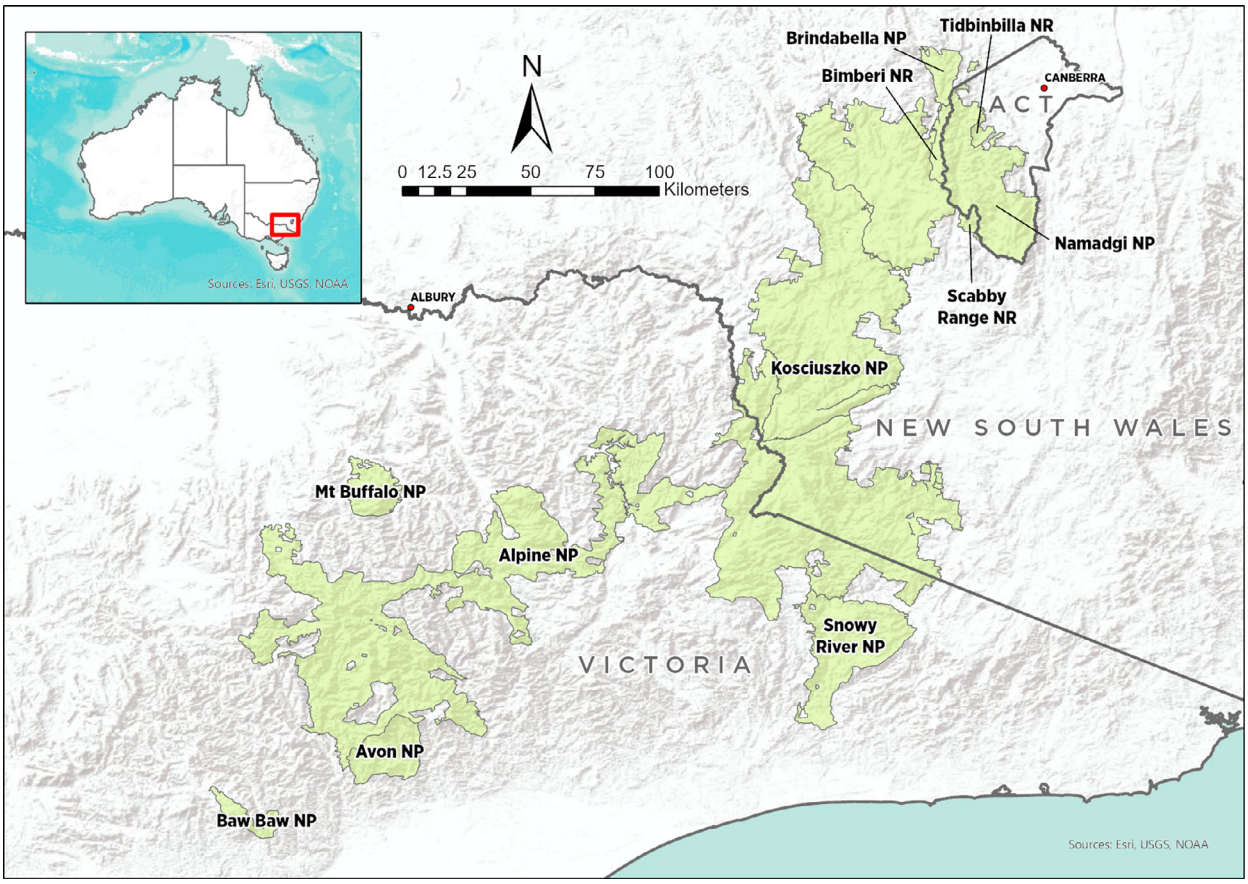



**Fig. 1.** The Australian Alps location.

than the steep, rocky environments of other alpine systems (Chang *et al.* 2015; Donhauser & Frey 2018).

Considerable research attention has been focused on a range of above-ground ecosystems in the Australian Alps, including numerous reviews, surveys and meta-analyses, relating to various aspects of flora and fauna (e.g. Spellerberg 1998; Hunter & Gillespie 1999; Costin *et al.* 2000; Johnston & Ryan 2000; Raadik & Kuiter 2002; Green & Osborne 2003; McDougall & Wright 2004; Scherrer & Pickering 2005, 2006; Bear *et al.* 2006; Pickering *et al.* 2007; Broome *et al.* 2013; Doherty *et al.* 2015; Mackey *et al.* 2015). However, comparatively little work has been conducted on the soils of the region, their nature, distribution and vulnerability to external pressures and threats. Our search for existing literature relating to the Australian Alps generated 277 items from 1985 to present relating to the physical environment, conservation, ecology and management of the Australian Alps. However, of these, only seven related specifically to soils. We acknowledge that this search did not include a range of 'grey literature' or literature generated prior to this date. Nevertheless, the result illustrates the relative paucity of research attention on the soil resource by comparison with other disciplines.

Soils are a fundamental component of all terrestrial ecosystems (Ritz 2011) and a fuller understanding of the role they play in the ecosystems of the Australian Alps is required to assist effective management strategies. This review was undertaken to: (i) summarise current knowledge relating to soils in the Australian Alps, including their nature, distribution and characteristics, and the services they provide; (ii) assess their vulnerability to external pressures and threats and (iii) recommend research priorities to guide on-ground and strategic management actions.

## KEY ASPECTS AND DISTRIBUTION OF SOILS IN THE AUSTRALIAN ALPS

### Geological setting

The Australian Alps were formed by early Cretaceous uplift (Müller *et al.* 2016) and comprise a range of parent materials, including Ordovician metasediments, Silurian granites and Devonian volcanics and

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sediments (Best *et al.* 1964; Adamson & Browne 1966; Adamson & Loudon 1966; Lewis & Glen 1995; Spate & Baker 2018). In the late Cretaceous and Palaeogene periods, basaltic lavas were also extruded in the Cabramurra-Kiandra area of NSW, and in Victoria, particularly around Mount Hotham (VandenBerg 1999).

Much of the NSW section of the Australian Alps retains the pre-uplift peneplain surface (Costin 1954), with steeper slopes associated mainly with the incised valleys of the Murray, Snowy, Thredbo and Tumut rivers. The Victorian Alps are more dissected, with only small areas of the original peneplain surface remaining at Bogong High Plains, Dinner Plain and Howitt Plains. The Australian Alps had very limited cirque glaciations during the Last Glacial Maximum (LGM) (Costin 1954), although periglacial activity was more widespread, with gravelly deposits (Barrows *et al.* 2004), solifluction terraces and non-sorted steps, formed by seasonal deep freezing and thawing (Costin *et al.* 1967, 1969).

## Soil data and mapping

Costin (1954) described the main soil types and vegetation alliances of the 'Monaro' region, defined to include Kosciuszko NP. Within the NSW section of the Australian Alps, 62 profiles were described with laboratory data, and a broad-scale soil map prepared. This remained the only soil mapping in the NSW Alps until reconnaissance mapping was undertaken as part of the Southern Comprehensive Regional Assessment (DLWC 1999). The western section of Kosciuszko NP (Murray River catchment) is also covered by reconnaissance mapping (OEH 2010) as is a small section in the north-western corner of Kosciuszko NP (B. R. Jenkins, unpublished data). Soil mapping in the NSW Alps is, therefore, largely at a broad-scale reconnaissance level and not at a scale suitable for soil and catchment management (Gunn *et al.* 1988; Schoknecht *et al.* 2008). Most of the region remains without site-based soil descriptions. Until recently, there were only eight profiles with laboratory soil data within the entire Kosciuszko NP. A further 47 alpine humus soils have recently been described and sampled by Oliver (unpubl.) and, in 2020/2021, the Yarrangobilly Soil and Landscape Assessment project described 90 soil profiles in the north-western corner of Kosciuszko NP with samples from 61 profiles subjected to laboratory analysis (Jenkins *et al.* unpubl.).

The Victorian section of the Australian Alps also has little soil information at scales better than reconnaissance, with the most comprehensive information in a series of 'land system' reports which integrate climate, geology, physiography and native vegetation, in addition to descriptions of typical soils. The mapping has been combined into a consistent State-wide map (Rowan 1990). A small amount of data have also been generated by universities (e.g. Rowe & Anderson 2006). Soil sampling was undertaken associated with 17 Terrestrial Ecosystem Research Network (TERN) sites established in the Australian Alps to monitor changes in biodiversity. However, the soil sampling methodology used did not meet national protocols (McKenzie *et al.* 2002; Bowman *et al.* 2009; Sanderman *et al.* 2010).

In the ACT, Talsma (1983) described profiles in the Cotter River catchment, with eight profiles located in the north-western montane section of Namadgi NP. The eastern section of Namadgi was mapped by Jenkins (1993), with the entire ACT portion of the Australian Alps mapped by Cook *et al.* (2016). Hydrogeological landscapes of the ACT were mapped by Muller *et al.* (2015). Soil profile descriptions for both the NSW and ACT sections of the Australian Alps are available from the NSW Soil and Land Information System (SALIS). In summary, however, the lack of a detailed understanding of soil properties and distribution is a fundamental limitation to our knowledge of the soils of the Australian Alps.

## Factors influencing soil distribution

In the seminal work '*Ecosystems of the Monaro Region*', Costin (1954) identified the major soil types of the Australian Alps. The principal soil types are summarised in Table 1 using the original classification of Hallsworth and Costin (1950), with the equivalent Australian Soil Classification (ASC) (Isbell & NCST 2021) provided. Table 1 also outlines the general climatic, physiographic and geologic settings, associated vegetation types and typical features of each soil. Type profiles and their typical locations in the landscape of the Australian Alps are also represented in Figure 2.

Geology, climate and topographic position are principal determinants in the development and distribution of soils worldwide, and these are also controlling factors in the Australian Alps. Soil types at lower elevations in the Australian Alps appear to be closely related to parent material. For example, the Podsolic Soils (Kandosols/Chromosols) of the tablelands and lower elevation montane zones (Costin 1954) are generally found on granite substrates, while Red Loams (Ferrosols) are found mainly on basalts. With increasing elevation, the influence of parent material on soil type diminishes, with precipitation, temperature and soil biological processes becoming more important (Costin 1955). Soil and vegetation patterns in the Australian Alps broadly correspond (Costin

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**Table 1.** Principal Soils of the Australian Alps (adapted from Costin 1954; Mason & Williams 2013)

| Great soil group (after Hallsworth & Costin, 1950) | Distribution | Climate | Physiography | Geology | Vegetation | Typical Features | SOC ranges (%) (Costin 1954, unless *= Oliver unpubl. data) | Equivalent Australian soil classification (Isbell and NCST, 2016) |
|---|---|---|---|---|---|---|---|---|
| Brown/ Grey-Brown Podsolic Soils | Widespread in tablelands and lower montane zones | Wide range of climatic conditions but typically 460–640 mm precipitation, including light winter snow. Mean monthly temperature range is 21° in mid-summer to 3° in mid-winter | Wide range of environments, except where drainage is impeded | Granites and sedimentary rocks | Dry tussock grassland and dry eucalypt forest/woodland | Brownish loam to sandy loam $A_1$ horizon topsoil, often with a pale $A_2$, overlying reddish to yellowish brown loam to clay loam subsoil. Slightly to moderately acid with relatively low organic matter content. Increasing sesquioxides and clay content with depth | Topsoils (A) 0.44–2.4 Subsoils (B) 0.10–4.6 | Grey/Brown Chromosols |
| Gley Podsols | Occurs in tablelands, montane, sub-alpine and alpine zones | Wide range of climatic conditions, 460–1270 mm precipitation, including light-to-heavy winter snow. Mean monthly temperature range is 21° in mid-summer to −5.3° in mid-winter | Level to moderately inclined slopes with a watertable in the subsoil | Most rock types except basalts and limestones | Wet tussock grassland and woodland, and heath | Greyish to black loam to clay loam $A_1$ horizon topsoil, often with a pale $A_2$, overlying yellowish or brownish clay loam subsoil. Moderately acid with relatively low organic matter content. Increasing sesquioxides with depth but with little increase in clay | Topsoils (A) 1.15–2.70 Subsoils (B) 0.21–0.46 | Yellow/Brown Kandosols |
| Fen Peats | Localised occurrence in tablelands, montane, sub-alpine and alpine zones | Wide range of climates, 460–1270 mm precipitation, including light to heavy winter snow. Mean monthly temperature range is 21° in mid-summer to −5.3° in mid-winter | Level and gently sloping areas, permanently wet | Largely independent of rock type | Fen (sedges), occasionally short alpine herbfield | Brownish black fibric, hemic and sapric peats, up to >1 m, overlying grey to gleyed alluvial/colluvial gravels to clays. Moderately to extremely acid peat overlying alluvium and/or gravels | Topsoils (F) 10.83–17.02 Subsoil (H) 3.07–10.67 Gravels 0.44 | Organosols |
| Red Loams | Restricted occurrence in eastern and north-western montane zone | Average annual precipitation 760–1400 mm, including light-to-moderate winter snow. Mean monthly temperature range is 18° in mid-summer to 2° in mid-winter | Wide range of environments except where drainage impeded | Typically basalt | Wet eucalypt forest | Dark-brown to reddish-brown clay loam A horizon topsoil, strong fine structure, grading to a reddish sub-plastic clay loam to light clay. Strongly to extremely acid, high percentages of sesquioxides | Topsoils (A) 4.36–5.13 Subsoils (C) 0.26–0.33 | Red Ferrosols |
| Meadow Soils | Widespread in tablelands and montane zones, with limited occurrence in the alpine zone (snow patches) | Wide range of climates, but typically 460–700 mm precipitation. Mean monthly temperature range is 25.2° in mid-summer to −1.0° in mid-winter | Level to gently inclined slopes. 'Snow patch' meadow soils are confined to the base of semi-permanent snow patches | Fine-textured alluvium and colluvium from a range of rock types | Wet tussock grassland eucalypt woodland and short alpine herbfield | Brownish black to black well-structured, organic clay loam to clay A horizon topsoil, becoming more mineral and lighter coloured, and with a watertable, and gley horizon with depth. Commonly underlain by a gravel bed at ~1 m depth. Acid to slightly acid, high clay contents. | Topsoils (A) 3.36–5.36 Subsoils (BG) 0.19–1.63 | Leptic Tenosols, Redoxic Hydrosols |

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**Table 1.** *Continued*

| Great soil group (after Hallsworth & Costin, 1950) | Distribution | Climate | Physiography | Geology | Vegetation | Typical Features | SOC ranges (%) (Costin 1954, unless *= Oliver unpubl. data) | Equivalent Australian soil classification (Isbell and NCST, 2016) |
|---|---|---|---|---|---|---|---|---|
| Transitional Alpine Humus Soils | Widespread in sub-alpine and upper montane zones, typically above 1300 m elevation | Average annual precipitation 760–1270 mm, including moderate winter snow. Mean monthly temperature range is 20.1° in mid-summer to −4.7° in mid-winter | Wide range of environments except where drainage impeded | All rock types except basalt and limestone | Wet, tall eucalypt forest. | Dark, organic A horizon topsoil, loams to clay loams, often with a greyish $A_2$, overlying C horizon of weathered rock. Less well-developed organo-mineral A horizon than the alpine humus soils. Moderately to strongly acid topsoils. Often more coarse fragments than alpine humus soils | Topsoils (A) 3.33–4.01 Subsoils ($A_3$, $A_4C$) 0.31–0.74 | Chernic Tenosols |
| Alpine Humus Soils | Widespread in alpine and sub-alpine zones, typically above 1500 m elevation | Average annual precipitation 760–2290 mm, mainly as persistent winter snow. Mean monthly temperature range is 14° in mid-summer to −5.3° in mid-winter | Wide range of physiography except where extremely windswept or impeded drainage | Independent of rock type | Tall alpine herbfield, sub-alpine eucalypt forest, tussock grassland and heath | Thick, dark, organic A horizon topsoil, loams to clay loams, strong fine structure and high porosity, gradual boundary change to C horizon of weathered rock. Strongly acid topsoils | Topsoils (A) 11.18–16.02 Topsoils ($A_1$) 3.53–8.89 2.92–28.01* Subsoils ($A_{3/4}C$) 0.34–2.90 | Chernic Tenosols |
| Lithosols | Limited occurrence throughout the region | Wide range of climates | Steep upper slopes, snow patches and exposed peaks and plateaux | Wide range of geological types | Found within a wide range of vegetation communities on windswept, exposed ridge crests of the Alps, often associated with feldmark vegetation | Shallow soils dominated by rock and coarse material. Properties dependent on parent material but have poor moisture status | No data | Clastic Rudosols |

Tableland zone (≤1000 m), Montane (1000–1500 m), Sub-alpine (1500–1800 m) and Alpine (1800 m+).

1954; Pickering & Green 2009; Kirkpatrick *et al.* 2014) (Table 1) because the same processes and factors that control the distribution of soils also control the distribution of vegetation communities, and both vegetation and soil systems are mutually dependent.

It is therefore useful to discuss the soil types of the Australian Alps in terms of biophysical zones based on the close association between vegetation communities and elevation. These zones are as follows: tablelands (≤1000 m); montane (1000–1500 m); subalpine (1500–1800 m; and alpine (1800 m+) (Good 1992; Doherty *et al.* 2015; Murphy & Fogarty 2019). At local scales, other factors may also include site exposure, site drainage, fire history and aeolian dust accessions (Costin 1954; Johnston 2001; Kirkpatrick *et al.* 2014).

## Key soil types of the Australian Alps

Certain soil types are confined to the Australian Alps and are, therefore, of particular scientific interest. Upland Peat Soils (Organosols) are rare on the Australian continent but are relatively common in the Alps, with 6037 ha of bogs and fens mapped within Kosciuszko NP (Whinam *et al.* 2003; Hope at al. 2012), 2700 ha of peatlands above 1000 m in Victoria (Lawrence *et al.* 2009) and 350 ha of peat soils in the ACT (Hope 2006). Peats are found in high rainfall areas (>1400 mm annually) and in permanently waterlogged locations which limit the breakdown of organic matter (Hope *et al.* 2012) with some peats up to 3.9 m deep (Kemp 1993). Peats deliver a range of significant ecosystem services, not least their critical role in the hydrology of the region, releasing water slowly and maintaining a constant supply downstream (Grover & Baldock 2013; Karis *et al.* 2016). In contrast, degraded peatlands retain less water, and release it quickly, causing erosion of rivers and siltation of reservoirs (Costin *et al.* 1959, 1960). Peat soils are also sensitive indicators of environmental change (Grover *et al.* 2012; Hope *et al.* 2012), and in their accumulated organic layers retain a

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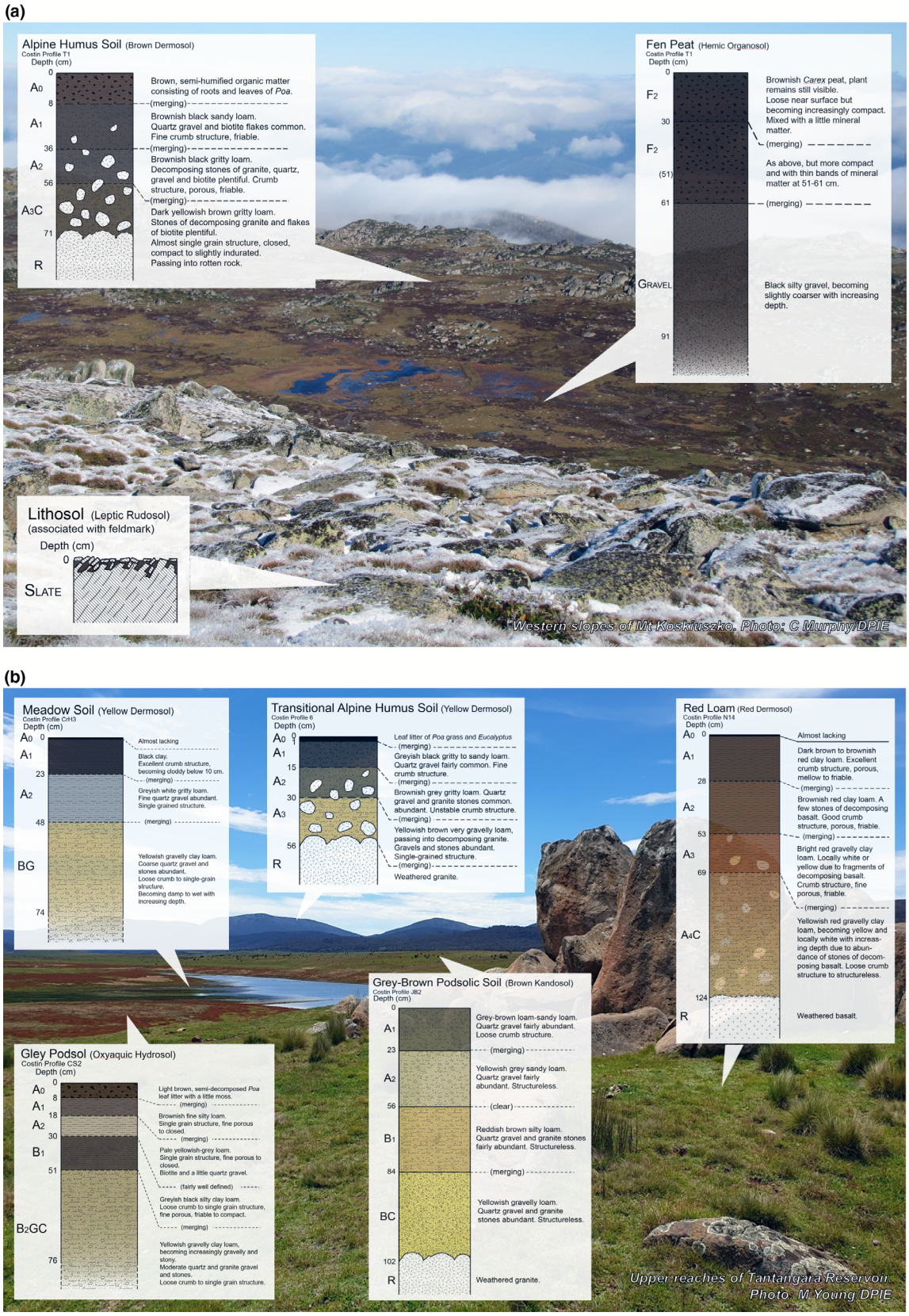



**Fig. 2.** Principal Soils of the Australian Alps (after Costin 1954): (a) High Elevation and (b) Low Elevation. Soil profile and horizon nomenclature and horizon descriptions are from Costin (1954).

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record of environmental, climate and land use change through time (Grover *et al.* 2012). Several peats in the region have been dated to the immediate post-LGM ~12 000 BP (Kemp 1993; Martin 1999; Hope *et al.* 2009, 2012).

Despite the importance of peat soils across the Australian Alps, our understanding of their rates of formation and particularly their vulnerability to change remains incomplete. Grover *et al.* (2005) noted, '*the importance of bogs in the catchment hydrology of the Australian Alps has been long recognised but little studied*' and many authors have also recommended further research (Grover *et al.* 2005; Hope *et al.* 2005, 2012; Good 2006; Good *et al.* 2010; Whinam *et al.* 2010a, b; Grover & Baldock 2012).

Alpine humus soils and transitional alpine humus soils (Chernic Tenosols) (Isbell & NCST 2021), with dark organic A horizons overlying weathered rock, are widespread on relatively gentle slopes across the upper montane, subalpine and alpine zones (Table 1). Costin (1955) suggested that the alpine humus soils of the Australian Alps are deeper and more fully developed compared to soils of other alpine zones due to their prolonged *in situ* soil development in a minimally glaciated environment. The largest occurrence of alpine humus soils is found in Kosciuszko NP, although their full spatial extent has not been accurately determined. These soils are particularly well structured, with a high water-holding capacity and, like the peats, are critical to the delivery of water to lower catchments (Costin *et al.* 1964; Worboys *et al.* 2011; NPWS 2014).

Several processes have been proposed to explain the formation and characteristics of transitional and alpine humus soils, although the precise mechanisms involved remain obscure. Brewer and Haldane (1973) noting the presence of stone lines and the sudden texture and mineralogical changes, suggested that these soils consist of two stratigraphic units. Costin (1954) and Wood (1970) also noted the abundance of earthworm bioturbation, including the mountain earthworm (aff. *Megascolex* sp). The significance of these species was demonstrated by Mitchell (1988) who found that at elevations >1200 m, the entire soil profile above the stone line was bioturbated. Costin *et al.* (1952) suggested that fine topsoil textures included colloidal material from the decomposition of snow-grasses, while Walker and Costin (1970) and Johnston (2001) proposed aeolian dust deposition as a factor in the formation of these soils. The formation of these soils undoubtedly results from a combination of such factors but further investigation is required to fully elucidate their mechanisms of formation.

Nevertheless, it remains true that there is a general inadequacy of information on alpine humus soils (Costin *et al.* 1952). Basic issues, such as the processes involved and rates of formation, and their vulnerability to external threats (including climate change) have received scant systematic research attention.

## ENVIRONMENTAL SERVICES PROVIDED BY SOILS IN THE AUSTRALIAN ALPS

### Water delivery

Alpine soils are key for the delivery and quality of water in the Australian Alps. Notwithstanding threats from feral animals and several land uses, the water generated from Australia's alpine areas is generally considered clean and reliable. The Australian Alps deliver approximately 29% of the Murray–Darling Basin's water. This is 9600 Gigalitres of water, worth an estimated $9.6 billion to the Australian economy (Worboys & Good 2011; Worboys 2015). This service is provided through several mechanisms. Soil impacts on water by influencing energy, storage, filtration, biogeochemical transformations and temperature. Water infiltration into soil significantly slows the movement of water, reducing the energy and erosive power of water as it progresses downslope. This results in a reduction in turbidity, flash flooding and stream bank, sheet and gully erosion. Water pollution from high suspended sediment concentrations can be further exacerbated by the material bound to the sediment, including heavy metals, pathogens and nutrients (Lintern *et al.* 2018).

Water is stored in alpine soils and the release of water occurs over a longer period than water derived from overland flow alone. Water is filtered through the soil and many impurities, fine sediments and salts are removed from the water. Soil allows for the biogeochemical transformations of phosphorus and nitrogen removing them from both surface and subsurface flows by adsorption (phosphorus) and denitrification (nitrogen) (Guo *et al.* 2019; Lintern *et al.* 2018). Water temperature is kept relatively constant in soils, ensuring that the temperature of water delivered to streams is also relatively constant (Mainali & Chang 2021). Soils in the Australian Alps, therefore, play a key role in the reliable delivery of clean water to lower catchment ecosystems and communities.

### Carbon storage in soils

Globally, soils are estimated to contain 1500 Pg of carbon, which is more than is contained in vegetation and the atmosphere combined (Batjes 1996; Houghton 2005; Lal & Follet 2009; Scharlemann *et al.* 2014). As such, soils represent a significant component of the global terrestrial carbon cycle, with the potential to act as sinks or sources. Soils with large

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quantities of soil organic carbon (SOC) are especially important, with relatively small changes in soil carbon (positive or negative) accounting for very large quantities of greenhouse gas (GHG) emissions. Knowledge and appropriate management of such soils therefore have considerable significance in the context of climate change and our response to it (Lal 2004).

The total quantity of carbon stored in the Australian Alps has not been quantified but many soils store large quantities of SOC (Table 1) due to their high elevation combined with the prevailing wet and cold climate. Modelling suggests that many higher elevation parts of the Australian Alps have mineral soil SOC stocks >200 tC $ha^{-1}$ (Gray *et al.* 2015, 2016) at 0–30 cm depth, significantly higher than the average soil organic carbon concentration (29.7 tC $ha^{-1}$) across the continent (Viscarra-Rossel *et al.* 2014). In contrast to most 'alpine' zones globally, the quantities of soil carbon increase with elevation in the Australian Alps (Costin 1955; Chang *et al.* 2015; Chen *et al.* 2016; Doolette *et al.* 2017).

Alpine humus soils in particular contain large concentrations of SOC (Costin 1954; Costin *et al.* 1952; McHugh 2016; Oliver unpubl. data; Table 1). However, apart from short-term studies on carbon fluxes over alpine humus soils (McHugh 2016) and peat soils (Gunawardhana *et al.* 2018), carbon accumulation rates, processes and total stock in these soils have received little attention.

Peat soils globally represent a significant component of the carbon cycle and are estimated to contain around 500 000 Tg C (Gorham 1991; Bergman *et al.* 2000; Yu *et al.* 2009; Gorham *et al.* 2012; Müller & Joos 2020). Peat soils store more SOC per ha than mineral soils, as they typically have higher carbon contents throughout the soil profile than their mineral counterparts. Hope and Nanson (2015) mapped peatland systems in the NSW Alps and found that they cover some 8000 ha (0.16% of the area) and store 49 million $m^3$ of peat with an average carbon stock of 444 tC $ha^{-1}$, which is a figure approached by few other landscapes in eastern Australia.

The impact of climate change on peat carbon stocks is an issue of growing global concern (Gorham *et al.* 2012; Qiu *et al.* 2020) and peats in the Alps are likely to be sensitive to the combined effects of warming and hydrological modification, with the potential for significant changes in the carbon they store (Hope & Nanson 2015). This appears to be occurring currently as Grover *et al.* (2005) and Grover and Baldock (2012) noted that many wet peats in the Australian Alps have degraded and dried, leading to carbon loss.

### *Biodiversity*

The biological richness of soils is understood to be orders of magnitude greater than the above-ground ecosystems that they support (Anderson 1995; Beare *et al.* 1995; Wall-Freckman *et al.* 1997; Orgiazzi *et al.* 2016) and soils significantly contribute to the biodiversity and ecosystem function. Soil organisms contribute to a wide range of ecosystem functions, including the decomposition of organic matter, nutrient cycling, bioturbation and the suppression of soil-borne diseases (Brussaard 1997; van der Heijden 2008).

### *Soil macro- and meso-biota*

The development of alpine humus soils was partly attributed by Costin (1954, 2004), Mitchell (1988) and Good (1992) to the decomposition and redistribution of plant remains by invertebrates, particularly earthworms. Soils of the Australian Alps appear to provide an unusually favourable habitat for earthworms (Wood 1974), and several cold-adapted species are found only in Australian alpine and sub-alpine environments (Jamieson 1973). These findings contrast with other alpine areas of the world where worm numbers tend to decrease with increasing elevation (Palm *et al.* 2013; Garcia-Franco *et al.* 2021).

It is likely that many of the abundant insect fauna in alpine and sub-alpine zones of the Australian Alps (some 850 recorded species: Good 1992) depend upon the soils of the region and their capacity to provide over-winter habitat (Costin 2004). Both insects and fungi play important roles in the processes of decomposition and cycling of organic matter and nutrients in soils. In a study of litter decomposition in a stand of mature snow gums (*Eucalyptus niphophila)* (Macauley 1975), fungi were involved in soil phosphorus (P) mineralisation, while insects were important in nitrogen (N) cycling. Johnston and Ryan (2000) found that almost all plants in alpine herbfields in Kosciuszko NP were colonised by arbuscular mycorrhizal fungi. This work suggests an inter-dependence between the soils, macro- and meso-fauna in the Australian Alps, but very little work has been done since to explore these relationships in more detail.

### *Soil microbial diversity*

Microbial processes are known to play a major role in soil development and are associated with virtually all nutrient and elemental cycles (van der Heijden *et al.* 2008; Schimel & Schaefer 2012; Rousk & Bengston 2014). It is also likely that much genetic and functional diversity lies within the soil microbial community (Cluzeau *et al.* 2012) as is often the case in mountain ecosystems (Rime *et al.* 2015; Donhauser & Frey 2018). However, in the Australian Alps, microbial diversity and functions have received relatively little research attention (Wunderlin *et al.* 2016). Soil microbial populations are typically

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controlled by elevation and climatic parameters, including temperature and precipitation (Donhauser & Frey 2018; Adamczyk *et al.* 2019) and these factors are likely to be of particular relevance for the Australian Alps and their soil microbial composition. The diversity and functionality of soil biota in the Alps are therefore also likely to be vulnerable under changing environmental conditions.

## PRESSURES AND THREATS TO SOILS IN THE AUSTRALIAN ALPS

In the last ~200 years, soils of the Australian Alps have been impacted by a range of new land uses, including grazing and burning (Breckwoldt 1988), works associated with the Snowy Mountains Scheme and ski resorts (McPhee & Wilks 2013; Buckley *et al.* 2000), as well as impacts from wildfire (Zylstra 2006), feral animals (Cairns 2019) and visitors (Pickering 2010). These threats continue across the Australian Alps.

### The legacy of domestic animal grazing

Much of the Australian Alps was originally characterised by nearly complete vegetation cover, with abundant organic material and leaf litter providing effective control of surface runoff and soil loss (Urbanska & Chambers 2002; Johnston 2003; Scherrer & Pickering 2005). Sheep and cattle were introduced to the open high plains of the Australian Alps by the 1830s and erosion associated with grazing and annual burning soon became a major issue (Helms 1893; Byles 1932; Stretton 1946; Gare 1968; Breckwoldt 1988; Young 1991). Grazing was progressively excluded from the NSW Alps from 1944 (Breckwoldt 1988; Good 2006), with a major driver being the establishment of the Snowy Mountains Scheme in 1949 and the economic need to manage catchments for water yield (Good 1992). It was not until 1972 that the remaining leases were terminated (Good 1992). Grazing also ceased in the 1970s in the area that was to become Namadgi NP in the ACT (AANP 2013). In Victoria, grazing was discontinued in Alpine NP in 2005 (AANP 2013).

Soils in the Australian Alps typically have a low resistance to trampling by hard-hooved herbivores and are highly susceptible to soil erosion (Costin 1954; Wahren *et al.* 1994; Good 1995; Newsome *et al.* 2002). Early damage was reported by Helms (1893), Byles (1932) and in more detail by Costin (1954). The NSW Soil Conservation Service commenced rehabilitation of 1550 ha of severely eroded alpine areas along the Main Range in 1963 (Good 1992; NPWS 2014), with works completed in 1982 (Worboys & Pickering 2002). In many areas, damage to soils and vegetation was so severe that erosion continued long after grazing was removed (Costin 1959; Totterdell & Neubauer 1973; Good 2006). Indeed, Johnston (2003) suggested that where soil erosion was extensive, plant cover might never re-establish without significant rehabilitation works (Scherrer & Pickering 2001; McDougall 2005; Good 2006). Even areas that appeared to have been successfully rehabilitated required repeated treatment (Irwin & Rogers 1986; Scherrer & Pickering 2001; Worboys & Pickering 2002).

Peat soils are particularly vulnerable to ongoing degradation, and impacts, including pugging, incision and drying, have been documented in NSW (Costin *et al.* 1959; Good 2004), Victoria (Lawrence *et al.* 2009) and the ACT (MacDonald 2009). It has been suggested that the incision of many wetlands and streams might be largely irreversible without active soil conservation measures (Good *et al.* 2010).

Restoration of soils in the Australian Alps is, however, challenging due to severe climatic conditions and short growing seasons. Furthermore, exposed soils are subject to frost heave (Good & Johnston 2019), where repeated freeze and thaw cycles exacerbate soil surface disturbance, necessitating rapid rehabilitation. Many surface soils are also typically low in nutrients (Costin 1954; MacPhee 2012; MacPhee & Wilks 2013), which further constrains recovery (Costin 1954; Scherrer & Pickering 2005). Restoration, therefore, needs to be implemented rapidly with ongoing use of restoration sites limited (Urbanska 1997; Bakker & Berendse 1999; McDougall 2001; Scherrer & Pickering 2006). Efforts to rehabilitate disturbed lands in the Australian Alps will require actions specifically designed for these challenging environments and significantly longer investment and time for recovery than might be expected elsewhere.

### Fire

Aboriginal fire management in the Australian Alps region has been considered by Flood (1980), Banks (1989) and Zylstra (2006), who concluded that lower elevation grasslands and grassy woodlands would probably have been burned from periodically to encourage new grass growth, to encourage important plants such as the yam daisy and attract animals such as kangaroos and wombats. Shrubby montane forests were probably burned with fires of lower intensity and frequency, while wet montane forests and subalpine areas were probably burned rarely (Mooney *et al.* 1997) to keep pathways open. Nevertheless, the frequency of fires is largely unknown (Banks 1989).

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There was an increase in fire frequency in tableland environments in the late Holocene, possibly coinciding with increasing Aboriginal populations and resource use (Portenga *et al.* 2016).

The arrival of European graziers and prospectors caused a significant increase in fire frequency and intensity in many parts of the Australian Alps, with consequent impacts on soils (Clarke 1860; Pryor 1939; Dodson *et al.* 1994; Mooney *et al.* 1997). Burning was used to clear ground, to encourage the growth of grasses and to dry out bog and swamp areas for access (Irwin & Rogers 1986). However, many forest types responded by changing from open climax communities to thick stands of post-disturbance, coloniser species (Hancock 1972; Flood 1980; Pulsford 1991; Wahren *et al.* 1994; Zylstra 2006) and increasing the fire risk further. The largest areas affected by fire burned in 1939, 2003, 2006 and 2020. In Victoria, the combined 2003 and 2006 fires covered practically all of the Victorian alpine national parks (Esplin *et al.* 2003; Parks Victoria 2016), with the 2020 fire again affecting the north-eastern Victorian Alps and Mount Hotham (Huf & Mclean 2020; IGEM 2020). Most of Kosciusko NP was impacted by either the 2003 or 2020 fires, and almost all of Namadgi NP was impacted by severe fire in 2003 with the southern section burning again in 2020 (House of Representatives 2003; Owens & O'Kane 2020).

Fire typically results in the reduction in the levels of soil organic matter (SOM) and long-term decline in nutrient pools (Kirkpatrick & Dickinson 1984) by oxidation, volatilisation and convection and allows leaching and erosion from a site (Raison 1980; Leitch *et al.* 1983; Thomas *et al.* 1999; Tulau & McInnes-Clarke 2015; Tulau *et al.* 2019, 2020). The magnitude of changes to catchment hydrology depends on a number of factors, including the severity of a fire, soil type, antecedent soil moisture and post-fire rainfall intensity and erosion (Yang *et al.* 2018). Following intense bushfires, runoff/infiltration ratios and peak stream flows typically increase (Brown 1972), resulting in sediment movement potentially several orders of magnitude larger than normal. This is especially true where severe fire is followed by rainfall events of high intensity and erosivity (Brown 1972; Good 1973; Yang *et al.* 2018). Low severity fires are generally considered to have a lesser impact on soils and catchment hydrology than intense bushfires (Tulau & McInnes-Clarke 2015), largely due to the larger retention of ground cover and soil organic matter. However, severe erosive events have been recorded in south-eastern Australia following even low severity prescribed burns (Cawson *et al.* 2012). Costin (1954), therefore, recommended that fire be excluded from virtually all of the alpine and sub-alpine zones, including alpine humus soils, as well as large parts of the montane zone. Fire should also be avoided on other particularly vulnerable soil types, including karst (Spate 1990) and peats (Good *et al.* 2010).

In Victoria, Rowe (1970) noted evidence of soil erosion and changes in bogs due to fire in Mount Buffalo NP. Brown (1972) also examined the hydrological impacts of the 1965 fire near Tumut and found significant increases in runoff and peak discharge and a thousand-fold increase in erosion rates. Similarly, Good (1973) found severe erosion due to heavy rainfall immediately following wildfires that occurred in 1973 in Kosciuszko NP. Following the 2003 wildfire in Kosciuszko NP, Smith and Dragovich (2008) measured post-fire erosion, and detected significant differences between burnt and unburnt slopes. Slope stability was modelled by URS (2006) to identify catchments that were more susceptible to erosion following bushfires. Significant impacts on water quality and hydrology of catchments in the ACT were seen following the 2003 wildfires (Rustomji & Hairsine 2006; White *et al.* 2006), and Good *et al.* (2010) reported a range of bog and fen sites damaged by the same fire.

This review did not locate any studies that specifically examined fire-induced SOC losses in the Australian Alps, although data collected recently by Oliver (unpubl.) suggest that the 2020 wildfires alone resulted in the loss of 19% of the SOC present prior to the fire. Similarly, there are no known studies on the impacts of fire on soil biodiversity in the Australian Alps. It is possible to predict the vulnerability of different soil types to fires of varying severities and to predict impacts on SOC, soil biodiversity, erosion and hydrology. The locations, magnitude and impacts of erosion can be estimated in near real-time after fire and storm events, including identifying risks to the public and infrastructure posed by post-wildfire mass movements (Nyman *et al.* 2011; Yang *et al.* 2017). There is an expectation that the risk of wildfire will increase into the future, particularly in the temperate and montane forest systems of south-eastern Australia (Bradstock 2010) and the need for information concerning the responses of soils to fire is pressing.

## Climate change

Climate modelling projections suggest a 1.5–3°C increase in both minimum and maximum temperatures in the Australian Alps by 2050 (against 1990 values) coupled with a 5–20% decline in precipitation (Hughes 2003; Tuteja *et al.* 2004; Chiew *et al.* 2008; Slatyer 2010; Murphy & Rawson 2011; Olson *et al.* 2016). However, the frequency of extreme weather effects (e.g. drought) is also projected to increase into

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the near (2020–2039) and far future (2060–2079) (Herold *et al.* 2018) with the intensity and erosivity of rainfall events in the region predicted to increase by 2–8% in these time periods (Zhu *et al.* 2020). These climatic changes are occurring particularly rapidly at higher elevations and above-average warming is predicted in the alpine zone (Beniston 2003; Kullman 2004; Olson *et al.* 2016). As a consequence, significant reductions are predicted in both the frequency and duration of snow cover in the Australian Alps (Whetton *et al.* 1996; Whetton 1998; Hennessy *et al.* 2003; Ji *et al.* 2017; Di Luca *et al.* 2018) and indeed significant changes in regional climate, including declining snow cover, have already been reported (Green & Pickering 2009).

These climatic trends will undoubtedly have profound impacts on ecosystems of the Australian Alps (DEC 2006; DECC 2008) but assessment of these has focused almost exclusively on plant and animal responses, with little consideration of potential changes in the soil resource on which they depend. Rises in ambient temperature will inevitably lead to increased soil temperature, which is likely to have widespread, cascading ecological effects (Scherrer & Pickering 2001). Given the significance of climate in determining the distribution of soils in the Australian Alps, soils are likely to be particularly sensitive to projected changes in climate.

Murphy and Rawson (2011) modelled climate impacts on a range of soil properties across NSW, predicting a significant decline in the organic matter status of soils in the Australian Alps as increased temperatures speed organic matter decomposition (*cf.* White-Monsant *et al.* 2017). However, the net balance between carbon loss and gain from primary productivity is currently unknown (Davidson & Janssens 2006). More recently, Gray and Bishop (2019) projected SOC losses to 2070 exceeding 20 tC $ha^{-1}$ (0–30 cm) in the southern alpine regions of NSW and Victoria. Climate change is expected to particularly impact the most carbon-rich peat soils (Organosols) (Grover & Baldock 2010, 2012), although the net effect on the more extensive alpine humus and transitional alpine humus soils is also likely to be significant.

Biotic activity in the soil is likely to change significantly under a warming, drying climate (Classen *et al.* 2015), where soil organic matter decomposition rates typically increase with increasing temperature. These changes undoubtedly make the organic matter contained within soils of the Australian Alps vulnerable to decomposition (Hersich *et al.* 2015; Guja & Brindley 2017). Donhauser and Frey (2018) reviewed the effects of climate change on the microbiology of alpine soils internationally and, while highlighting likely shifts in soil community structure and function, concluded that they are under-studied. In relation to the Australian Alps, Pickering and Green (2009) concluded that little or no information is available in the literature on the possible responses of alpine soil fauna to climate change.

Following reports of deaths of snow gums (*E. pauciflora*) in Kosciuszko NP in 2008, 'dieback' has expanded rapidly in recent years across the Australian Alps (McDougall *et al.* 2018). Very little is known about snow gum dieback, and while it has been linked to the native longicorn beetle (*Phorocantha* sp.), climate change-induced soil moisture stress has also been implicated (Scanlon 2019). It is therefore expected that impacts on snow gums will increase as climate change proceeds, but our lack of knowledge of soil-related factors limits our ability to predict the significance and extent of this tree decline.

## Impacts from tourism, infrastructure and pollution

Increasing visitor numbers in the Australian Alps has intensified impacts on the soil resource in recent decades. Impacts are associated with tourism (including the ski industry), infrastructure (including roads) and activities such as walking, mountain biking and horse riding. Ballantyne and Pickering (2015) reviewed research relating to visitor impacts on alpine areas worldwide and found a range of soil-related impacts, including soil exposure, nutrient enrichment, compaction and erosion (Delgado *et al.* 2007; Freppaz *et al.* 2013). Further work since that date has similarly reported soil compaction associated with ski pistes/slopes (Allegrezza *et al.* 2017; Pintaldi *et al.* 2017). Many such impacts occur in the Australian Alps (Keane *et al.* 1979; Cousins 1998; Pickering *et al.* 2003; Scherrer & Pickering 2006) and these impacts are likely to intensify as visitor numbers continue to grow (Buckley *et al.* 2000; Worboys & Pickering 2002; Growcock 2005). Soil impacts have been reported at the six ski resorts in NSW and five in Victoria (Cousins 1998; Pickering *et al.* 2003; NGH Environmental 2007) and Pickering *et al.* (2003) found that, although spatially limited, these impacts can extend to adjacent natural areas. Soil impacts might also result from linear infrastructure such as power lines but we could find no published research.

Roads can have significant effects on the soil resource. Pickering and Hill (2007) found that road verges in Kosciuszko NP were more gravelly and acid compared with soils away from the roads, with some depositional areas affected by sediment and nutrients. A further concern is the impact on soils of de-icing salts applied to roads (Amrhein *et al.* 1992). Roads within Kosciuszko NP were de-iced with salt (NaCl) until 1999, after which calcium chloride ($CaCl_2$) has

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been used (Hocking 2010). High application rates were identified along a section of Kosciuszko Road with significant impacts on soil chemistry, structure and nutrients (Allen 1999; Johnston & Johnston 2004; Hocking 2010; Findlay & Kelly 2011), although the magnitude, extent and impact of deicing salts more broadly remain largely unquantified. Erosion on unsealed roads such as fire trails can be significant and the design, maintenance and soil type from which trails are constructed are important determinants of erosion hazard (Croke and Mockler 1997; NSW Rural Fire Service 2016; Soil Conservation Service of NSW 2017).

Hiking and mountain biking can result in vegetation loss, soil exposure and compaction and changes in hydrology (Pickering *et al.* 2010, 2011). Scherrer and Growcock (2006) examined off-track walking impacts in the Kosciuszko summit area and demonstrated long-term soil damage after only moderate usage levels. The importance of mitigating visitor damage to vegetation and soils has resulted in programmes aimed at minimising further damage (Parr-Smith & Polley 1998; Pickering *et al.* 2003) and restoring damaged sites (Good & Grenier 1994; Pickering *et al.* 2003). Horse riding, driving off-road vehicles and the addition of nutrients (e.g. by detergents, faecal material and urine) were also seen as major, visitor-related issues (Worboys & Pickering 2002; Pickering *et al.* 2003).

The major works phase of the Snowy Mountains Scheme was completed in 1979 (Good 1992) but the legacy of scheme works has been an ongoing issue. Rehabilitation has occurred at over 200 sites within Kosciusko NP, with a further 100 sites identified (MacPhee & Wilks 2013). The majority of these sites comprise areas of crushed, excavated rock or sites disturbed by infrastructure. However, asbestos contamination is also present at some former Snowy Scheme sites, including Island Bend, where contaminated material has been capped (Wilks 2019). The Snowy 2.0 project, approved in 2020, involves works, including excavations, road upgrading and other construction, covering ~105 ha in total (EMM Consulting 2018).

The legacy of mining also remains, with contamination of soil, sediment and water by metals such as copper, zinc, nickel and arsenic, at locations such as the Lobs Hole copper mine in Kosciuszko NP (EMM Consulting 2018). There are 28 former mining fields recognised in the Victorian Alps and 40 in the NSW Alps, mostly for gold, but also copper, silver, lead, tin, chromite, tungsten and molybdenum (Kaufman 2002). Early rehabilitation works to address erosion from the grazing era involved the use of galvanised wire netting, which has led localised zinc toxicity and plant death (Johnston 1995; Johnston & Good 1996). The wire netting was eventually replaced (Good 2006) but the long-term effects of such pollution on soils remains unclear.

Cloud seeding is a potential and widespread soil contamination source. Silver iodide (AgI) has been used since 2004 as a seeding agent to increase precipitation across the main range of Kosciuszko NP. Williams and Denholm (2009) reviewed the literature relating to AgI and, although adverse ecosystem effects appeared to be negligible, they noted the limited extent of monitoring. This remains a potential concern, as the bioavailability of silver is controlled by pH and redox status, and could represent a pollution hazard in the wet, acid soils of the Australian Alps.

## Feral animals

An emerging issue in the Australian Alps is the impact of introduced large animals, principally horses and deer (Worboys 2016; AAS 2018). Cairns (2019) estimated that there were >25 000 feral horses in the Australian Alps, with 66% of these in NSW. Although drought and the 2020 wildfires may have reduced this number, the longer-term trend is an annual increase of 23% (Cairns 2019). The negative impacts of feral horses have been well documented in both the NSW and Victorian Alps, and include erosion and pollution from trampling, pugging, grazing, track creation, collapsing stream banks and incision of wetlands and streams (Dyring 1990, 1992; Ward-Jones *et al.* 2019; Dawson 2009; Dawson & Axford 2011; Wild & Poll 2012; Worboys & Pulsford 2013; Robertson *et al.* 2015; Tolsma & Shannon 2018; AAS 2018). Peat soils are particularly susceptible to damage by horses, causing channelised flow, degrading water-storing capacity and redistributing nutrients (Costin 1954; Dyring 1992; Dawson 2009; Pickering *et al.* 2010; Cleary 2013). Driscoll *et al.* (2019) concluded that extensively damaged bog communities could take millennia to recover.

The more extensive soil types across the Australian Alps such as alpine humus soils and transitional alpine humus soils are also susceptible to damage by horses (Costin 1954; Good 1995; Newsome *et al.* 2002; Eldridge *et al.* 2020) due to loss of surface litter and cryptogams, increased compaction and reduced aeration, water infiltration, pore space and soil water (Dyring 1990, 1992; Yates *et al.* 2000). Horse faeces and urine contain high concentrations of N and P and the addition of manure can lead to local nutrient hotspots along tracks and trails. In areas of high horse densities, nutrients can affect vegetation, favouring species adapted to higher nutrients leading to the potential proliferation of weed species

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(Pickering *et al.* 2010). Under some circumstances, soil disturbance by hooved animals can create conditions suitable for pathogen invasion, and soil movement has been implicated in transferring *Phytophthora cinnamomi* and associated root pathogens from infected to non-infected areas (Newsome *et al.* 2002).

Deer species (principally fallow, red and sambar) have also formed wild populations that are growing and expanding their range across the region and increasingly becoming a management issue (Deer Advisory Council of Victoria 1979; Dunn 1985; Moriarty 2004; Forsyth 2006; Davis *et al.* 2016). In 2019, the deer population in the Australian Alps was estimated to be 7630 with an annual increase of 27% (Cairns 2019). Davis *et al.* (2016) reviewed our knowledge of deer numbers and distribution across Australia and proposed a series of research priorities, including population monitoring and interactions with native flora and fauna. Internationally, evidence suggests that soil degradation is caused by deer at high population densities (Mohr & Topp 2001; Harrison & Bardgett 2004; McDowell *et al.* 2004; Shelton *et al.* 2014). In these international environments, however, deer species are typically native and the extent and impacts of feral deer on soil properties in the Australian Alps are unclear. Other large introduced herbivores, including cattle, goats and pigs, have been observed across the Australian Alps (Cairns 2019) and rabbit infestations resulting in severe erosion occur locally (Leigh *et al.* 1987) but again the effects of these animals on the broader soil resource has received scant attention to date.

## RESEARCH PRIORITIES RELATING TO SOILS IN THE AUSTRALIAN ALPS

This review has highlighted a range of issues and threats that exist to the soil resource in the Australian Alps. Management of the Australian Alps national parks is carried out primarily by State and Territory agencies, with the Australian Alps Liaison Committee (AALC) facilitating cross-border co-operation. The AALC broadly recognises catchment and ecological issues and the threats from climate change as priority issues and encourages research. The Kosciuszko National Park Plan of Management (PoM) (NPWS 2014), for example, includes a number of actions relating to matters identified in this review, including soil and landscape mapping, soil erosion and rehabilitation, incision of drainage lines, rehabilitation of former work sites and mining areas, soil pollution and cloud seeding.

However, it is acknowledged that knowledge gaps exist, and that these need to be informed by research (NPWS 2014). NPWS (2014) recognised that the soils of the park have been little studied compared with most other elements of the landscape, and that *'this paucity of knowledge impacts upon the ability of the Service to formulate management regimes designed to maintain or enhance the long-term integrity of the soils of the park'*. More recently, detailed research priorities for the management of the NSW Alps were developed by NPWS (Wilson *et al.* 2018). Table 2 includes those related to soils, along with existing research priorities in the PoM.

The priority action relating to soils was a soil and landscape survey, including the identification of high value and vulnerable soils and the collection of baseline data. The current soil descriptions and mapping in the Australian Alps are not adequate to support current and future management needs. This work was described as '*an essential first step which will underpin all future soil research in the park*' (Wilson *et al.* 2018) and the availability of high-resolution remote imagery and spatial data (including LiDAR) has revolutionised what can be achieved in the area of soil and geomorphic mapping (Boettinger *et al.* 2010; Shi *et al.* 2012). The lack of well-established monitoring sites across the Alps with appropriate protocols is a significant weakness in our capacity to track and quantify changes in the soils and is a priority need to support future management actions.

The impact of climate change on the soils of the Australian Alps is the least understood threat, and filling knowledge gaps in this area is a high priority (Wilson *et al.* 2018), including the effects of climate change on SOC stability and flux, soil biota, soil structure and erosion rates. Climate change is also related to the incidence and impacts of fire, and wildfire is considered to have the greatest potential for widespread impacts on soils in the short to medium term (NPWS 2014), including its impact on SOC and catchment hydrology. Soil disturbance and erosion associated with fuel reduction burns are also of concern. The Kosciuszko NP PoM proposed research into the effects of various fire management regimes and practices on soils, landforms and catchment hydrology and stability. This work was seen as essential for the Fire Management Plan, which aims to ensure that fire-sensitive soils and landforms are protected from the damaging effects of fire and that catchment stability and water quality are not impaired. Assessment of post-fire landscape risks and mass movement hazards, especially associated with resorts, roads and trails, are also priorities.

One of the key emerging issues and knowledge gaps across the Australian Alps is the impacts of feral animals, principally horses and deer. The monitoring of these are priorities under the Australian Alps Cooperative Management Program (AALC 2018, 2019), and existing and predicted extent, abundance and impact of large introduced herbivores on soils

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**Table 2.** Priority research needs for management

| Issue/Knowledge gap | Recommendation |
|---|---|
| Soil mapping and data | • Carry out a detailed soil survey to better understand the nature and spatial distribution of soils, identify high value and vulnerable soils and collect baseline data.<br>• Collect reference data and establish a soil monitoring programme using appropriate protocols to measure and predict likely changes in the soil resource. |
| Alpine humus soils | • Research to better understand the processes and rates of formation of alpine humus soils and transitional alpine humus soils, and their vulnerability to external threats, including climate change, and their contribution to increased $CO_2$ emissions. |
| Peat soils | • Research to understand the processes and rates of formation of peat soils, and their vulnerability to external threats, including climate change, and their contribution to increased GHG emissions.<br>• Conduct a risk assessment of peat soils, and further research and establish a programme to restore peat soils. |
| Climate change | • Identify key soils potentially vulnerable to climate change, and their spatial extent, and quantify carbon accumulation rates and processes, total stocks and the stability of carbon and its vulnerability to climate change.<br>• Research the impact of warming on soil pathogens.<br>• Improve our understanding of the role of soils in snow gum dieback. |
| Soil biota | • Undertake an assessment of alpine soil biota to identify the locations of greatest value, richness, uniqueness or vulnerability.<br>• Research into the roles of soil microbes and fauna into soil formation and processes, and the possible responses of soil biota to climate change. |
| Domestic animal grazing | • Assess the rehabilitation of lands degraded by erosion. |
| Fire | • Research the impacts of wildfire and prescribed fire on soils, including SOC, soil structure, soil biodiversity and functions, landforms and catchment hydrology.<br>• Undertake post-fire remedial works on a range of bog and fen sites damaged by recent wildfires, to prevent further loss by erosion.<br>• Assess post-fire landscape and mass movement hazards, including risks to the public and infrastructure. |
| Visitors and infrastructure | • Research the impacts of increased visitation, particularly on vulnerable soils.<br>• Assess the impacts of biking and hiking trails for different soils and landscape elements, including steep slopes and saturated slopes.<br>• Evaluate the long-term success of restoration efforts undertaken in response to damage from Snowy Hydro and visitor facilities, including soil condition, health and function. |
| Feral animals | • Research the impact of large mammals, particularly horses and deer, on degradation of peats and other soil types, erosion, nutrients and spread of soil pathogens.<br>• Examine the potential role of feral animals in the movement of pathogens, including *Phytophthora cinnamomi*. |
| Soil pollution | • Investigate long-term impact of silver iodine from cloud seeding and other contamination sources on soils.<br>• Research the impact of de-icing salts to investigate the short- and long-term impacts on salts on soils and ecosystems, streams (water quality and bank erosion) and vulnerable ecosystems (e.g. bogs and fens) and investigate alternatives. |

and biota is a key research need under the Greater Alpine National Parks Management Plan (Parks Victoria 2016). The Alpine NP has a specific feral horse strategic action plan (Parks Victoria 2017), which aims to monitor feral horse numbers, and assess stream, vegetation and habitat condition, as does the Kosciuszko NP draft Wild Horse Management Plan (NPWS 2016). Assessing and managing increasing visitors' impacts from biking, hiking and skiing is also considered a high priority by park managers. The research priorities identified are limited to those we deem to be the most important. The order in which they are addressed will be dependent on the priorities set by park management and resource availability.

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